

Microsoft MarkItDown + Claude Desktop

Setup Summary Notes

Source: github.com/microsoft/markitdown · June 2026

★ Key Advantages of MarkItDown + Claude Desktop

✓ All points below are verified against the official [microsoft/markitdown](#) and [markitdown-mcp](#) README files on GitHub.

1. Token-Efficient Format — LLMs Natively Speak Markdown

The official README states that mainstream LLMs "have been trained on vast amounts of Markdown-formatted text" and that "Markdown conventions are also highly token-efficient." This means Claude processes Markdown reliably while using fewer tokens than raw PDF text or HTML — leaving more context for reasoning.

2. Document Structure Is Preserved

The README explicitly states MarkItDown focuses on "preserving important document structure and content as Markdown (including: headings, lists, tables, links, etc.)" Raw PDF extraction often collapses tables and scrambles columns. MarkItDown keeps structure intact, so Claude sees a clean, accurate representation of your document.

3. Broad File Format Support — Built-in

Directly from the official README, supported formats include: PDF, Word (.docx), Excel (.xlsx/.xls), PowerPoint (.pptx), HTML, CSV, JSON, XML, ZIP (iterates contents), Images (OCR), Audio (speech transcription), YouTube URLs, and EPub. All convert to Markdown Claude can read natively mid-conversation.

4. One Tool, Any URI — Seamless via MCP

The MCP README states the server "exposes one tool: `convert_to_markdown(uri)`, where `uri` can be any `http:`, `https:`, `file:`, or `data:` URI." Claude calls this automatically during conversation — no manual scripts, no copy-pasting, no pre-processing pipeline required.

**URI — Uniform Resource Identifier : Any kind of resource identifier (URL, Local resources - File path)*

6. Local Execution — Trusted & Private by Design

The MCP README states it is "meant for local use, with local trusted agents" and "binds to localhost by default." Running via Docker further sandboxes the process with no filesystem access unless explicitly mounted. Your documents are converted on your own machine — no content is sent to a third-party service.

Source: [microsoft/markitdown-mcp](#) README

1. What Is MarkItDown?

MarkItDown is Microsoft's open-source Python tool that converts documents into clean Markdown so Claude can read and process them natively. It connects to Claude Desktop via the Model Context Protocol (MCP).

Built by: Microsoft AutoGen Team (Adam Fourney, adamfo@microsoft.com)

License: MIT

Package: mcp/markitdown (Docker) · markitdown-mcp (PyPI)

Requires: Python ≥ 3.10 · Claude Desktop

2. How It Works

Once set up, Claude gains a tool called `convert_to_markdown(uri)`. When you give Claude a file or URL, it automatically calls this tool, converts the content to Markdown, and reads it — no manual conversion needed.

Supported File Formats

Microsoft Office: DOCX, XLSX, PPTX, XLS

PDF: PDF

Images (OCR): JPG, PNG, GIF, BMP, TIFF

Audio (transcript): MP3, WAV, M4A

Web: HTML, HTM

Data: CSV, JSON, XML

Archives: ZIP (iterates contents)

Text: TXT, MD, RST

3. Setup — Docker Method (Official Recommendation)

Microsoft's own PyPI docs state: "It is recommended to use the Docker image when running the MCP server for Claude Desktop." Docker runs the server in a fully isolated container.

Prerequisites

- Claude Desktop → claude.ai/download
- Docker Desktop → docker.com/products/docker-desktop (must be running before Claude starts)

Step 1 — Pull the Official Image

```
docker pull mcp/markitdown
```

Step 2 — Edit the Config File

- Windows: %APPDATA%\Claude\claude_desktop_config.json
- macOS: ~/Library/Application Support/Claude/claude_desktop_config.json

From claude desktop – Settings -> Developer -> Edit config

Paste this exact JSON into the file:

```
{
  "mcpServers": {
    "markitdown": {
      "command": "docker",
      "args": ["run", "--rm", "-i", "mcp/markitdown"]
    }
  }
}
```

Step 3 — Fully Quit and Reopen Claude Desktop

Closing the window is NOT enough. Right-click the Claude icon in the system tray → Quit Claude → Reopen.

Step 4 — Verify

Under developer section you should see convert_to_markdown listed under markitdown.

Optional — Mount a Local Folder

To let Claude read files from a local directory:

```
"args": ["run", "--rm", "-i",
         "-v", "C:/Users/YourName/Docs:/workdir",
         "mcp/markitdown"]
```

Then ask Claude: "Convert file:///workdir/report.pdf to markdown"

4. Security Best Practices

- Use STDIO mode only (default). No network port is opened.
- Never use HTTP/SSE mode for Claude Desktop — it has no authentication.
- Docker is safer than pip — fully isolated container.
- Only mount directories you intend to share.
- Install only from the official PyPI package: `pip install markitdown-mcp`
- Keep updated: `docker pull mcp/markitdown` or `pip install --upgrade markitdown-mcp`

5. Troubleshooting

- ▶ **Tool not showing after restart:** Docker Desktop wasn't running — start it first, then restart Claude.
- ▶ **JSON error on restart:** Validate: Get-Content
"\$env:APPDATA\Claude\claude_desktop_config.json" | python -m json.tool
- ▶ **Window close didn't work:** Must fully quit via system tray → right-click → Quit Claude.
- ▶ **Check MCP logs:** Get-Content "\$env:APPDATA\Claude\logs\mcp*.log" -Tail 30

6. Key Links

GitHub (official): github.com/microsoft/markitdown

PyPI package: pypi.org/project/markitdown-mcp/

Docker image: hub.docker.com/r/mcp/markitdown

Claude Desktop: claude.ai/download

Docker Desktop: docker.com/products/docker-desktop

Status: Setup confirmed working (tool verified in Claude Desktop) · June 2026